

2D Fast-Slow SBL Model

Physics & Parameterization (Atmospheric Science)

The model captures non-linear coupling between stable boundary-layer wind shear and turbulent kinetic energy (TKE).

- **Fast Variable e (TKE, m^2/s^2):** Relaxes on timescale $\epsilon \ll 1$. Its balance consists of shear production $\mathcal{P}$, regularized buoyancy flux $\mathcal{B}(e)$, and viscous dissipation ϵ_{diss} :

$$\epsilon \frac{de}{dt} = \mathcal{P} - \mathcal{B}(e) - \epsilon_{\text{diss}} = l_0 e S^2 - B_{0,\text{max}} \left(\frac{e^2}{e^2 + \delta_{\text{reg}}^2} \right) - \frac{e^3}{l_0 (1 + \beta N^2 / S^2)}$$

- **Hill-Type Saturating Regularization (δ_{reg}):** Rather than a polynomial expansion, $\mathcal{B}(e)$ is a Hill-type rational function (Hill coefficient $n = 2$). Near $e \rightarrow 0$, $\mathcal{B}(e) \approx B_{0,\text{max}} \frac{e^2}{\delta_{\text{reg}}^2} + \mathcal{O}(e^4)$, yielding an exact zero-TKE equilibrium with fast repelling linearized growth $\left. \frac{\partial \dot{e}}{\partial e} \right|_{e=0} = \frac{l_0 S^2}{\epsilon} > 0$ for $S > 0$ (modeling shear-driven turbulent reignition). For $e \gg \delta_{\text{reg}}$, it smoothly saturates to the physical cap $B_{0,\text{max}}$.
- **Slow Variable S (Wind Shear, s^{-1}):** Driven by synoptic geostrophic forcing G_0 , offset by turbulent shear stress extraction $\gamma_s e S$ and background relaxation $r_s S$:

$$\frac{dS}{dt} = G_0 - \gamma_s e S - r_s S$$

Fast-Slow Dynamical System (Mathematics)

The system is expressed in singular perturbation form $(\dot{e}, \dot{S}) = (\frac{1}{\epsilon} F(e, S), g(e, S))$.

- **Critical Manifold ($\mathcal{C}_0$):** The algebraic zero locus of the fast vector field:

$$\mathcal{C}_0 = \{ (e, S) \in \mathbb{R}_{\geq 0}^2 \mid F(e, S; \delta_{\text{reg}}) = 0 \}$$

- **Normal Hyperbolicity ($\mathcal{C}_0^+$):** The manifold sheet is normally attracting where the scalar fast eigenvalue is strictly negative:

$$\lambda_{\text{fast}} = \frac{1}{\epsilon} \frac{\partial F}{\partial e} < 0$$

- Saddle-Node Fold Boundary ($\partial\mathcal{C}_0$):** Point of normal hyperbolicity loss ($\lambda_{\text{fast}} = 0$) where stable turbulence collapses into the laminar state, defined by the 2D root system:

$$F(e, S; \delta_{\text{reg}}) = 0 \quad \text{and} \quad \frac{\partial F}{\partial e}(e, S; \delta_{\text{reg}}) = 0$$

Implementation Architecture

Theoretical Concept	Mathematical Formulation	Code Realization (<code>src/gspt_saddle_node_trajectory.jl</code>)
Critical Fold Boundary	$F(e, S) = 0, \frac{\partial F}{\partial e} = 0$	2D Newton-Raphson scheme in <code>solve_regularized_fold()</code>
Fast Subsystem Stability	$\lambda_{\text{fast}} = \frac{1}{\epsilon} \frac{\partial F}{\partial e}$	Scalar partial derivative evaluation in <code>fast_manifold_jac()</code>
Scalar Manifold Distance	$d_C^* = \frac{\ F(e,S)\ }{\max\big(1, \ \frac{\partial F}{\partial e}\ e + \ \frac{\partial F}{\partial S}\ S\big)}$	Step-by-step absolute value scalar metric tracking trajectory distance to $\mathcal{C}_0$
Partitioned Integration	$e_{n+1} = \max\big(e_n + \frac{\Delta t}{\epsilon} F(e_n, S_n), e_{\text{floor}}\big)$	$S_{n+1} = \max\big(S_n + \Delta t g(e_n, S_n), 0\big)$